



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY

ICT 1209-WEB TECHNOLOGIES
FINAL PROJECT REPORT

FLAVORVAULT

Theme:Recipe Book

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Submission Date: 2026/09/27

TABLE OF CONTENTS

1. Project Overview	3
1.1 Project Overview	3
1.2 Purpose	3
1.3 Target Audience.....	3
1.4 Key Features	3
 2. System Design.....	 4
2.1 Database Schema	4
2.2 Folder Structure	4
 3. Implementation	 5
3.1 Home Page	5
3.2 User Authentication (Login & Register).....	6
3.3 Contact Form	6
3.4 Recipe Browsing and Management	7
3.5 Responsive Design	8
 4. Challenges and Solutions.....	 8
 5. Individual Contribution.....	 9
 6. Setup Instructions	 9
 7. References	 10

1. PROJECT OVERVIEW

1.1 Project Overview

FlavourVault is a database-driven web application developed as a culinary and recipe sharing platform . The main purpose of the system is to provide a secure and user-friendly digital repository where food enthusiasts, home cooks, and professional chefs can discover, save, organize, and share their favorite culinary recipes . Instead of searching for cooking resources across fragmented sources, users can centralize their recipe management through FlavourVault . The application is designed with a simple and user-friendly interface to make culinary resources easier to access.

1.2 Purpose

The purpose of FlavourVault is to develop a web-based system that helps food enthusiasts and home cooks organize and access culinary recipes more easily . The system provides a centralized platform for accessing recipe materials and communicating with the administrator.

1.3 Target Audience

The main target users of the FlavourVault application include:

- Home cooks and culinary beginners looking for daily meal inspiration.
- Food bloggers and culinary enthusiasts wanting to archive and share recipes .
- General web users seeking an accessible, structured digital cookbook.

1.4 Key Features

The main features of the FlavourVault application are:

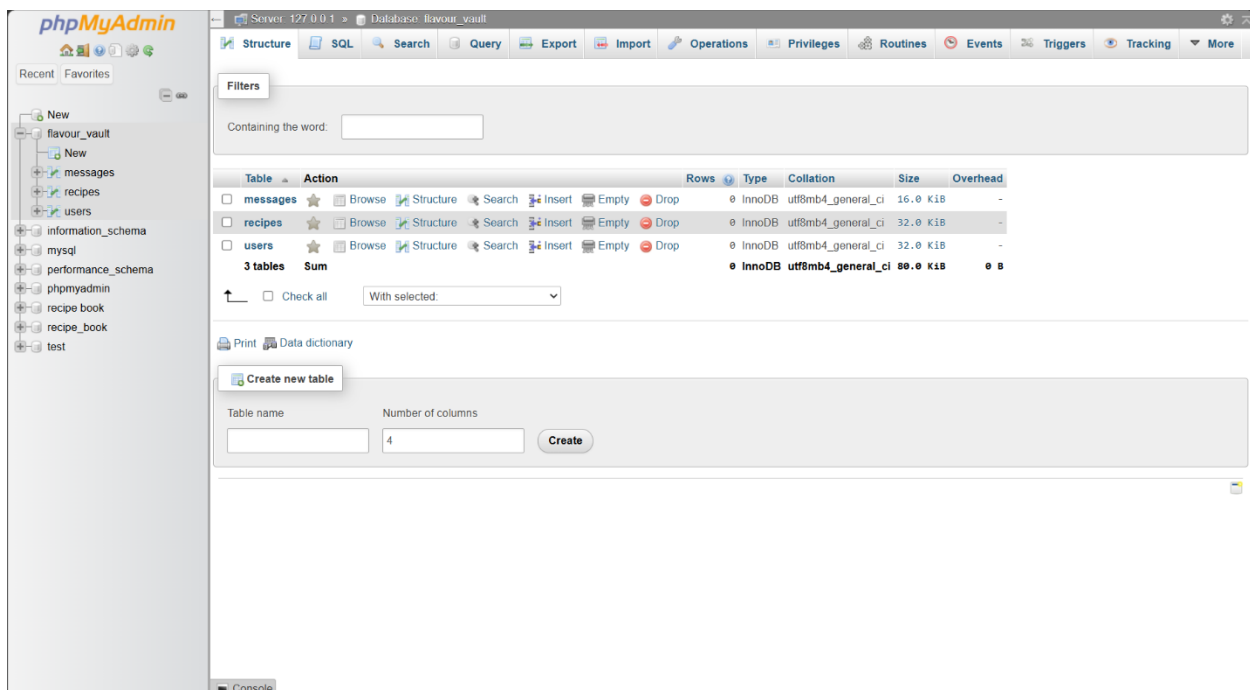
- **User Registration and Login:** Users can create an account, log in using their credentials, and log out from the system .
- **Recipe Management:** Users can view, browse, add, and update recipes according to different categories, preparation time, and serving sizes .
- **Search Function:** Users can search for recipes and culinary resources using a title name or keyword .
- **Contact Form:** Users can send inquiries, suggestions, or feedback to the administrator.
- **Responsive Design:** The website is designed to be usable on desktop computers, tablets, and mobile phones.

2. SYSTEM DESIGN

2.1 Database Schema

FlavourVault uses PHP and MySQL (PDO) for database-related processing . The supplied backend code confirms a users table used for login and registration, a recipes table used for recipe management, and a messages table used for contact-form submissions .

The login backend searches the users table using the email field and verifies the stored password. The contact backend inserts the submitted name, email, and message into the messages table.



2.2 Folder Structure

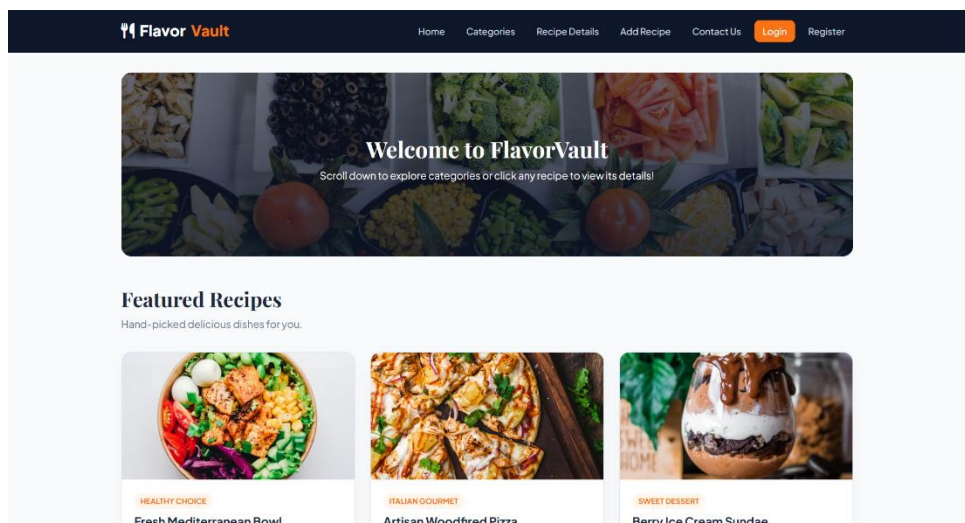
The project is organized into PHP/HTML files, CSS files, an image folder and authentication folders. The backend logic handles database connections, user authentication, and contact-message processing.

```
flavour_vault/  
├── auth/  
│   ├── login.php  
│   └── register.php  
├── css/  
│   └── style.css  
├── js/  
│   └── script.js  
├── dashboard.php  
├── index.php  
├── contact.php  
├── function.php  
└── flavour_vault.sql
```

3. IMPLEMENTATION

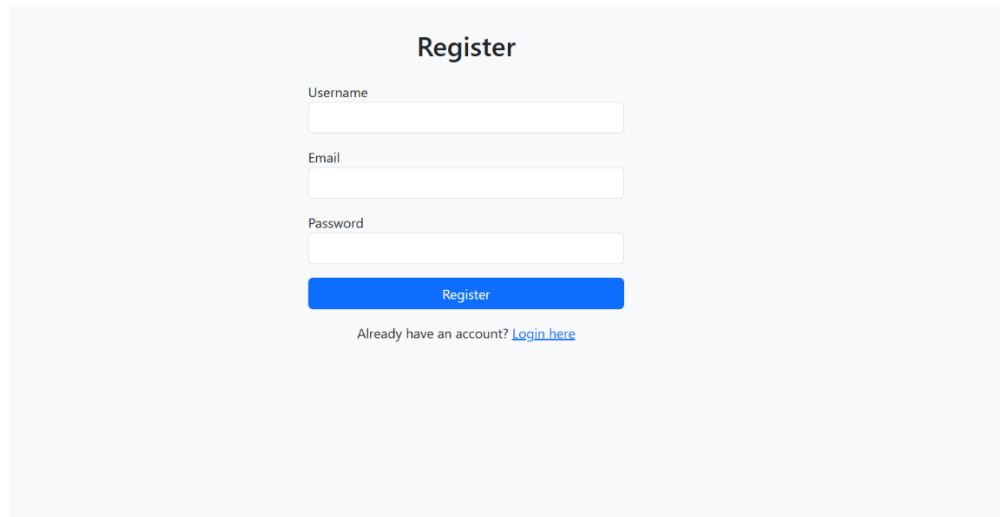
3.1 Home Page

The home page is implemented in index.php. It contains navigation links, welcome content, and an Explore Resources action . It also provides culinary categories including recipe collections and featured dishes .

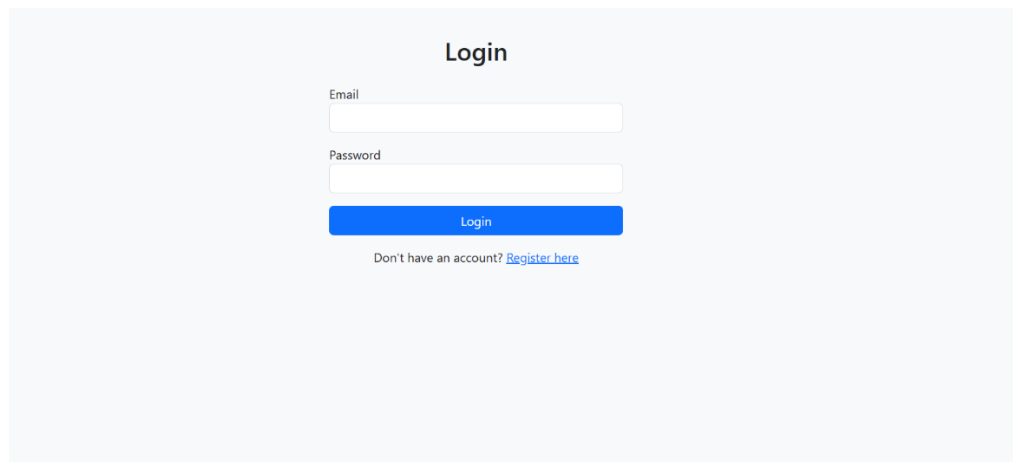


3.2 User Login & Register

The login page is implemented in login.php and register.php. It provides email and password fields and sends the submitted data to backend processing scripts using POST. The PHP backend uses prepared SQL statements to find the user by email and uses password_verify to check the submitted password.



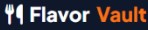
The Register form is centered on a light gray background. It features a title "Register" in bold black text. Below the title are three input fields: "Username", "Email", and "Password", each with a light gray border and a small label above it. A blue button with the text "Register" is positioned below the password field. At the bottom, there is a link that says "Already have an account? [Login here](#)".



The Login form is centered on a light gray background. It features a title "Login" in bold black text. Below the title are two input fields: "Email" and "Password", each with a light gray border and a small label above it. A blue button with the text "Login" is positioned below the password field. At the bottom, there is a link that says "Don't have an account? [Register here](#)".

3.3 Contact Form

The contact page provides fields for name, email, and message. The form sends the submitted data to backend processing scripts. The backend uses a prepared statement to insert the values into the messages table.


[Home](#)
[Categories](#)
[Recipe Details](#)
[Add Recipe](#)
[Contact Us](#)
[Login](#)
[Register](#)

Contact Us

Have questions or feedback? Send us a message below.

Name

Email

Message

SUBMIT

3.4 Recipe Browsing and Management

The home page presents different recipe categories and collections. The resource section is designed to display recipe information and search functionalities for users.


[Home](#)
[Categories](#)
[Recipe Details](#)
[Add Recipe](#)
[Contact Us](#)
[Login](#)
[Register](#)

Browse Recipes by Category

Filter dishes by meal type.

All
Breakfast
Lunch
Dinner
Dessert



HEALTHY CHOICE

Fresh Mediterranean Bowl

⌚ 15 mins prep



ITALIAN GOURMET

Artisan Woodfired Pizza

⌚ 30 mins prep



SWEET DESSERT

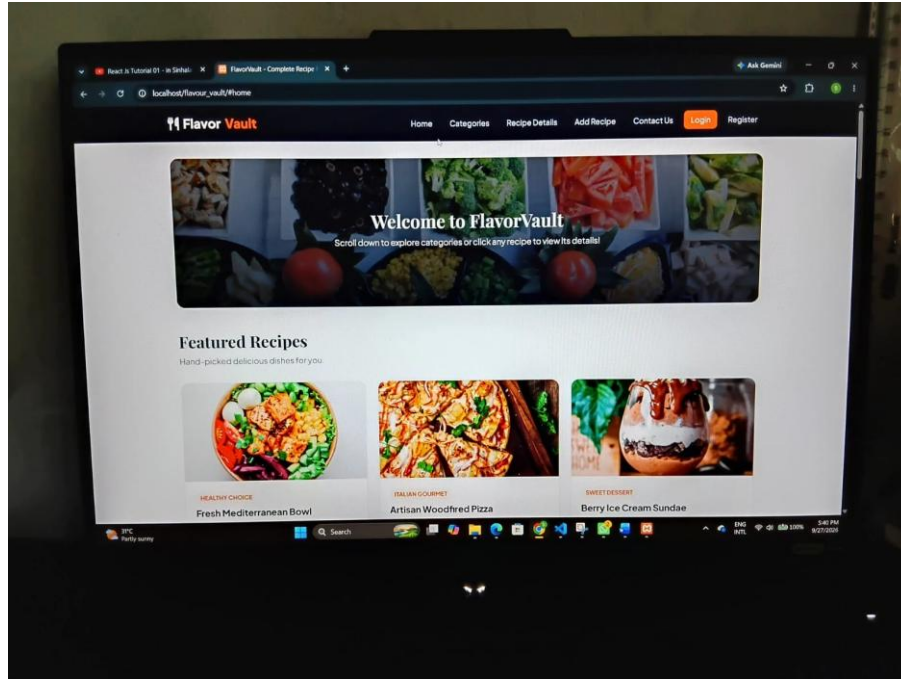
Berry Ice Cream Sundae

⌚ 10 mins prep



3.5 Responsive Design

The CSS files include media queries for smaller screen sizes. The home page adjusts its navigation, welcome section, recipe layout, and footer for smaller screens. The login and contact pages also contain responsive rules.



4. CHALLENGES AND SOLUTIONS

Challenge	Solution
Database Column Mismatches (Column not found)	Encountered SQL exceptions due to missing relational mapping (user_id foreign key) inside the recipes table. Resolved by modifying the table structure via phpMyAdmin .
Connecting Forms to Backend	Forms were configured with POST actions pointing to PHP backend files for processing .
Password Verification	The login backend uses password_verify to compare submitted passwords with stored hashed values .
Responsive Layout	CSS media queries were added to adapt the page layout for smaller mobile and tablet screen sizes .

Git Version Control Conflicts	Resolved branch merging discrepancies during collaborative feature integration using clean synchronization strategies .
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5. INDIVIDUAL CONTRIBUTION

Group Member	Individual Contribution
P.A.S.V.S.Lakshika	- Implemented backend logic (PHP/PDO database connections, authentication handlers, data retrieval queries) . - Designed core database schema architecture and performed system debugging . - Prepared and organized the final project report documentation.
W.K.M.Nikeshani	- Led frontend UI/UX design (HTML5/CSS3 layout structuring, responsive styling, navigation design). - Designed home page layouts and integrated contact and viewing interfaces. - Collaborated on system testing, debugging, and overall project improvements.

6. SETUP INSTRUCTIONS

To execute the FlavourVault project locally in a development environment using XAMPP:

1. Project Placement: Place the project directory flavour_vault inside your XAMPP installation root folder (C:\xampp\htdocs\flavour_vault).
2. Start Local Server: Open the XAMPP Control Panel and start both the Apache web server and MySQL database service.
3. Import Database:
 - Open your web browser and navigate to <http://localhost/phpmyadmin/>.
 - Create a new database with the exact name flavour_vault.
 - Select the database, go to the Import tab, and upload/execute the provided flavour_vault.sql file.
4. Launch Application:** Open a browser window and go to http://localhost/flavour_vault/index.php

7. REFERENCES

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- Nixon, R. (2021). *Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5* (6th ed.). O'Reilly Media.
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- W3C Web Standards Guidelines. Available at: <https://www.w3.org/> .